

$$G(s) = \frac{1}{(s+1)(s+2)}$$

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$$Y(s) = G(s) U(s) = \frac{1}{(s+1)(s+2)} \frac{1}{s}$$

$$E(s) = \frac{s^2 + 3s + 1}{(s + 1)(s + 2)s} = \frac{A}{s} + \frac{B}{s + 1} + \frac{C}{s + 2}$$

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$$A = \frac{1}{(1)(2)} = \frac{1}{2}$$

$$B = \frac{(-1)^2 + 3(-1) + 1}{(-1+2)(-1)} = 1$$

$$C = \frac{(-2)^2 + 3(-2) + 1}{(-2+1)(-2)} = -\frac{1}{2}$$

$$E(s) = \frac{0,5}{s} + \frac{1}{s+1} - \frac{0,5}{s+2}$$

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$$t \rightarrow \infty \rightarrow e(t) \rightarrow 0,5$$

$$e_{\infty} = 0,5$$

$$e_{ss} = 0,5$$

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$$e_{\infty} = 0,5$$

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$$e_{\infty} = A = sE(s)|_{s=0} = \frac{s^2 + 3s + 1}{(s+1)(s+2)}|_{s=0}$$

$$e_{\infty} = \frac{1}{2}$$

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Valor final

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$$

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$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$$

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{s^2 + 3s + 1}{(s+1)(s+2)} = \frac{1}{2}$$

$$G(s) = \frac{2}{(s+1)(s+2)}$$

$$E(s) = U(s) - Y(s) = \frac{1}{s} - \frac{2}{(s+1)(s+2)} \frac{1}{s}$$

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$$E(s) = U(s) - Y(s) = \frac{1}{s} - \frac{2}{(s+1)(s+2)} \frac{1}{s}$$

$$E(s) = \frac{s^2 + 3s + 2 - 2}{(s+1)(s+2)s} = \frac{s^2 + 3s}{(s+1)(s+2)s} = \frac{s(s+3)}{(s+1)(s+2)s}$$

$$E(s) = \frac{(s+3)}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$e(t) = A e^{-t} + B e^{-2t} \quad \rightarrow \quad e_{\infty} = 0$$

$$Y(s)$$



$$E(s)$$



$$e_{\infty}$$

$$Y(s)$$



$$y_{\infty}$$



$$e_{\infty}$$

$$G(s) = \frac{2}{(s+1)(s+2)}$$

$$Y(s)$$



$$Y(s) = G(s) U(s) = \frac{2}{(s+1)(s+2)} \frac{1}{s}$$

$$y_{\infty}$$



$$y_{\infty} = \lim_{s \rightarrow 0} \frac{2}{(s+1)(s+2)} = 1$$

$$e_{\infty}$$

$$e_{\infty} = 1 - y_{\infty} = 0$$

$$G(s) = \frac{s^2 + 6s + 9}{s^3 + 8s^2 + 17s + 10}$$

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$$y_{\infty} = 0,9$$



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$$Y(s) = G(s) \frac{1}{s} = \frac{G(s)}{s}$$

$$y_{\infty} = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s \frac{G(s)}{s} = \lim_{s \rightarrow 0} G(s)$$

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$$y_{\infty} = \lim_{s \rightarrow 0} G(s) = \frac{9}{10} = 0,9$$

$$u(t)=1$$

$$u(t)=5$$

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Sistema Linear

$$G(s) = \frac{s^2 + 29s + 208}{s^3 + 6s^2 + 10s + 208}$$

$$y_{\infty} = 10 \quad e_{\infty} = 0$$

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SAÍDA DIVERGE!

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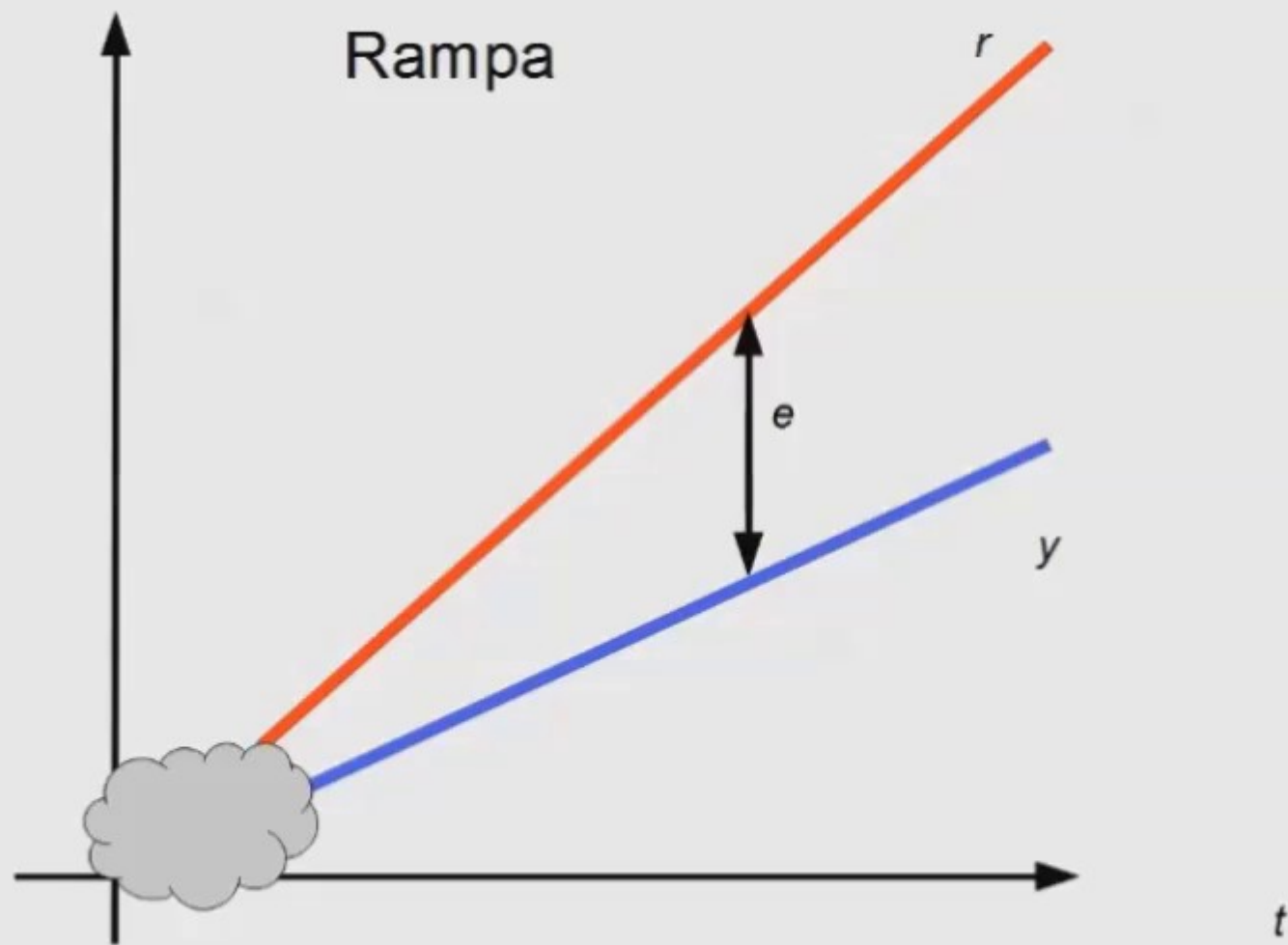
$$1 - G(s) = \frac{s^2 + 3s + 1}{(s+1)(s+2)}$$

$$E(s) = \frac{s^2 + 3s + 1}{s^2(s+1)(s+2)}$$

$$E(s) = \frac{s^2 + 3s + 1}{s^2(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1} + \frac{D}{s+2}$$

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$$e(t) = 0,5t + A + C e^{-t} + D e^{-2t}$$



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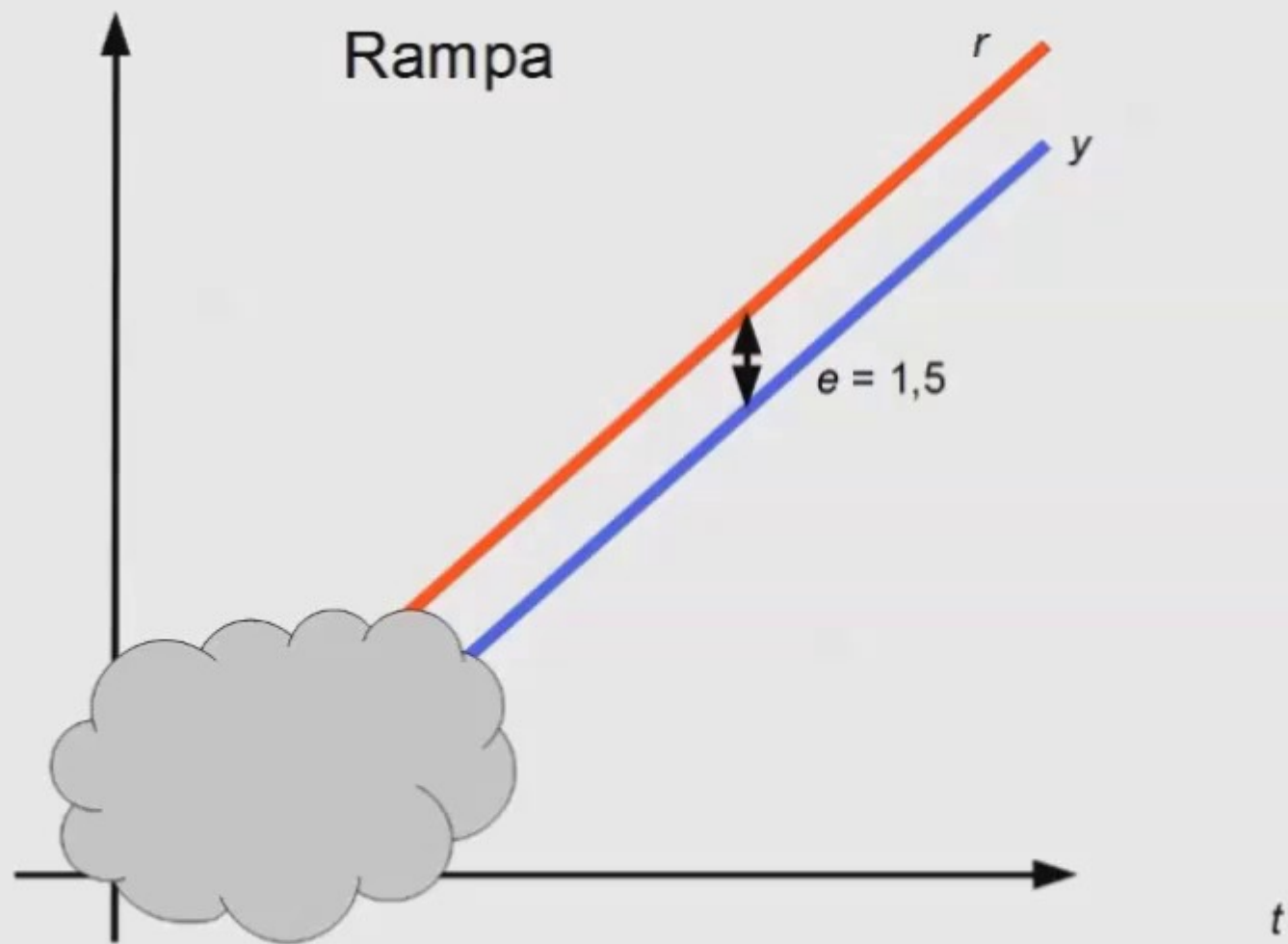
$$1 - G(s) = \frac{s^2 + 3s}{(s+1)(s+2)}$$

$$E(s) = \frac{s^2 + 3s}{s^2(s+1)(s+2)}$$

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$$e(t) = 1,5 + B e^{-t} + C e^{-2t}$$



$$G(s) = \frac{as^2 + bs + c}{s^3 + ds^2 + es + f}$$

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$$1 - G(s) = \frac{s^3 + ds^2 + es + f - as^2 - bs - c}{s^3 + ds^2 + es + f}$$

$$1 - G(s) = \frac{s^3 + (d - a)s^2 + (e - b)s + f - c}{s^3 + ds^2 + es + f}$$

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$$E(s) = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s - p_1} + \frac{E}{s - p_2} + \frac{F}{s - p_3}$$

$$E(s) = \frac{s^3 + (d - a)s^2 + (e - b)s}{s^3(s^3 + ds^2 + es + f)}$$

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$$E(s) = \frac{s + (d - a)}{s(s^3 + ds^2 + es + f)}$$

$$E(s) = \frac{A}{s} + \frac{D}{s - p_1} + \frac{E}{s - p_2} + \frac{F}{s - p_3}$$

$$e(t) = \frac{d - a}{f} + D e^{p_1 t} + E e^{p_2 t} + F e^{p_3 t}$$

$$G(s) = \frac{as^2 + bs + c}{s^3 + ds^2 + es + f}$$

$$c=f \quad b=e \quad \longrightarrow \quad e_{\infty(\text{parábola})} = \frac{d-a}{f}$$

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$$e_{\infty(\text{degrau})} = \frac{f-c}{f}$$

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